

Game Overview

Title: Hollow Head

Genre: 2D Arena Action Roguelite (Boss-Focused)

Platform: PC (Keyboard + optional controller support)

Engine: Godot

Target Audience: Hardcore Gamers

Core Gameplay Loop:

- Fight the boss -> beat the boss -> get an upgrade -> move to the next boss -> repeat

Mechanics:

Rules/systems:

- Player Upgrades
- Upgrades stay between levels
- Melee Attacks
 - Primary slash (fast)
 - Charged heavy attack
 - Directional input determines attack orientation
 - Light hit-stop and visual feedback on impact
- Dash / Dodge
 - Short invulnerability window
 - 2-directional
- 5 Health points
- 3 Difficulty levels (hearts regenerate between bosses, don't regenerate and 1 heart)
- 3 Bosses with their own boss rooms
 - Melee
 - Ranged
 - Brute

Dynamics:

- Expected Strategies
 - Possible upgrade paths the player might want to take
 - Attacks:
 - High damage
 - High attack speed
 - Mobility build:
 - Faster movement
 - Reduced dodge timer
 - Sustain build:
 - More hearts
- Emergent Interactions
 - Different upgrade synergies
 - Better at strategizing with the bosses
- Difficulty curve
 - Each boss get harder
 - Each boss has a new mechanic

- Different difficulty levels available to the player

Aesthetics:

- Fantasy World
- Level design will be based on fantasy tropes

Content plan:

- 3 boss arenas
- 3 boss levels
- 10-12 possible upgrades, roughly 3 choices per level
- Main menu
- Pause and restart system
- Win screen and game-over screen
- UI elements:
 - Health bar
 - Timer

Implementation plan:

- **Core Systems**
 - Player finite state machine (Idle / Attack / Dash / Hurt / Dead)
 - Boss phase state machine
 - Menu/Level State machine
 - Modular upgrade modifier system
 - Wave spawning manager
 - Stamina resource controller
 - UI manager
 - Scene reset system for clean restart
- **Technical Risks**
 - Upgrade stacking creating balance issues
 - Boss phase transition bugs
 - Incomplete state reset on restart
 - Performance drops during high projectile density
- **Mitigation Strategies**
 - Build upgrade system as modular effects
 - Develop boss in isolated test scene
 - Use scene reload for full state reset
 - Cap projectile counts and test early